Question:

A logic function is defined by the following minterms: $F(A, B, C) = \Sigma m(1,2,5,7)$. Express the function F in its canonical Product-of-Sums (POS) form. Simplify the expression obtained either using Boolean theorems or by using Karnaugh Maps. Write the function F in canonical Sum-of-Products (SOP) form also.

Answer

Convert each minterm into a product term.

Minterm	A B C	Product term
m1	001	$A'B'C$
m2	010	$A'BC'$
m5	101	$AB'C$
m7	111	ABC

Hence,

$$\boxed{F = A'B'C + A'BC' + AB'C + ABC}$$

This is the **canonical SOP**.

Step 2: Write the Canonical POS Form

The function is **0** for the remaining minterms:

$$m(0, 3, 4, 6)$$

Hence,

$$F = \Pi M(0, 3, 4, 6)$$

Now write each maxterm.

Maxterm	A B C	Maxterm
M0	000	$(A + B + C)$
M3	011	$(A + B' + C')$
M4	100	$(A' + B + C)$
M6	110	$(A' + B' + C)$

Therefore,

$$\boxed{F = (A + B + C)(A + B' + C')(A' + B + C)(A' + B' + C)}$$

This is the **canonical POS**.

3. Simplified SOP

$$\boxed{F = B'C + AC + A'BC'}$$

4. Simplified POS

$$\boxed{F = (A' + C)(B + C)(A + B' + C')}$$